

ADITI GUPTA

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EDUCATION

VIT Bhopal University, Bhopal, India | B.Tech in Computer Science & Engineering | CGPA: 8.83/10

Sep 2023 – May 2027

SKILLS

Languages: Python, SQL, C, C++

Machine Learning: Data Preprocessing, Feature Engineering, Dimensionality Reduction (PCA), Time Series Analysis, Ensemble Modeling, Model Evaluation & Validation

Frameworks & Libraries: scikit-learn, TensorFlow, Keras, Pandas, NumPy, XGBoost, LightGBM, CatBoost, Flask

Databases: MySQL, MongoDB

Tools & Platforms: Git, GitHub, Docker, REST APIs, VS Code, Jupyter

Familiar with: AWS, LLM Fundamentals

PROJECTS

Wake Drive | AI Driver Drowsiness & Cognitive Load Detection System

2024 – 2025

GitHub: github.com/aditi532/Cognitive-Awareness-Engine-for-Safe-Driving

- Enabled sub-second drowsiness detection on live video, measured by real-time flagging of microsleep events, by building a Flutter mobile app that computes Eye Aspect Ratio (EAR) from front-camera frames using Google ML Kit Face Detection.
- Reduced missed fatigue events by designing a three-stage escalation pipeline (UI warning at 0.5s, audio alarm at 1.0s, automated emergency trigger at 5.0s of unresponsiveness) with thresholds tuned through on-device testing.
- Automated SOS dispatch with zero manual input by integrating the Telegram Bot API to transmit live GPS coordinates and nearby hospitals retrieved via the OpenStreetMap/Overpass API.
- Sustained uninterrupted real-time inference on physical Android devices by engineering a custom NV21 frame converter and a high-contrast Medical Blue UI optimized for day and night driving conditions.

Brain Tumor Signal Classification | ML/DL Ensemble Model

Aug 2024

GitHub: github.com/aditi532/Brain-Tumor-Signal-Classification

- Classified diagnostic brain signals into 5 balanced categories at ~94% accuracy across 11,500 samples with 178 numerical features, by building a stacked ensemble (CatBoost + LightGBM base learners, Logistic Regression meta-learner) in Python and scikit-learn.
- Compressed the feature space from 178 features to 50 PCA components and a top-100 recursive-elimination subset, reducing model training cost while preserving classification accuracy.
- Established stacking as the most robust approach, measured by confusion-matrix and per-class performance across all 5 classes, by benchmarking it against Random Forest, XGBoost, and a 128→64→32 ANN with Dropout + L2 regularization.

AnomalyNet | Unsupervised Deep Learning Framework for Time Series Anomaly Detection

Aug 2024

GitHub: github.com/aditi532/AnomalyNet

- Detected anomalies in streaming time series with zero labeled examples, validated on the NAB benchmark dataset, by designing an unsupervised framework that scores points via MSE-based reconstruction and prediction error.
- Identified the strongest architecture for sparse-label, class-imbalanced data by implementing and benchmarking 4 models (MLP, 1D Conv AE, Variational 1D Conv AE, WaveNet) across both online and seq2seq settings.
- Cut manual tuning effort across experiment runs by engineering an automated hyperparameter grid search with JSON-based config management and metrics aggregation using Python (TensorFlow/Keras, NumPy, Pandas).
- Maintained detection accuracy under concept drift by implementing online training that continuously adapts the model to distribution-shifted streaming data.

CERTIFICATIONS

AI/ML Internship | MPOOnline Limited

May 2026 – Jul 2026

Advanced Software Engineering & Development Internship | MPOOnline Limited

May 2026 – Jul 2026

Applied Machine Learning in Python | Coursera

Nov 2025

AWS Technical Essentials | AWS Training & Certification

Sep 2025

EXTRACURRICULAR

Hackathons & Competitions | VIT Bhopal University

Aug 2023 – Present

- Cleared the college round of Smart India Hackathon (SIH), competing among top student teams, alongside participation in multiple other hackathons.
- Solved 200+ LeetCode problems in data structures and algorithms, focusing on consistency and problem-solving speed.